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The Editors-in-Chief
Advances in Structural Engineering
Sage

Dear Editors,

Submission of an original research article: “Observability and sensor layout for influence-area load monitoring of continuous girder bridges”

We submit the above manuscript for consideration as an Original Research Article in *Advances in Structural Engineering*. It falls within the journal’s focus on structural health monitoring for performance assessment of bridges.

Monitoring the loads that a bridge actually carries is often done with few channels, by integrating one response record and dividing by a calibration constant. On continuous girders this constant can vanish, because influence lines change sign over the piers. The manuscript gives structural engineers criteria for choosing, before installation, channels and positions that observe the load, and for judging whether their constant can be trusted when the pier stiffness is uncertain. The underlying identity and the generalized least-squares estimator are classical, and the manuscript says so; the contributions are the criteria built on them:

1. a closed-form catalog of blind loci for five channel types on prismatic two- and three-span girders, with a strain criterion that also holds for haunched girders and indicators that need no reference rigidity, showing that strain gauges are blind at every zero of the uniform-load moment and nearly blind in a band that can contain gauges set back from the piers;

2. a placement rule from extreme-point optimal design: for position-independent noise, optimal sensors lie at the extremes of the uniform-load response. On one day of public moving-truck records, weighting gauges by their vehicle-free noise covariance raised the out-of-sample error above that of a plain gauge average;

3. an identifiability analysis under uncertain pier stiffness. For a symmetric girder with equal piers, the constant follows a linear-fractional (Möbius) law. On an in-service 55 + 90 + 55 m box girder, static acceptance data leave its side-span deflection constant

uncertain by a factor of up to 6.6 but bound the main-span constant within a factor of 1.5, and a pier tiltmeter is the most informative additional measurement.

A laboratory test is consistent with the identity. The field records are used to check the assumptions behind the criteria; they cannot test weight discrimination, because their reference weights vary little, and the manuscript states this. The moving-load test of the in-service bridge is designed but not yet performed, and its predicted outcomes are given so that they can be checked.

The manuscript builds on bridge weigh-in-motion work published in this journal, including the review by Yu et al. (2016), and cites recent studies in the journal on vehicle-weight identification (Hao et al., 2019; Zuo et al., 2019) and on influence-line identification and model updating (Lin et al., 2023; Wang et al., 2026). It is about 11,600 word-equivalents with 7 figures and 4 tables; derivations, full field tables and the error budget are in the Supplementary Material.

Data availability. The field strain records are from the public NOR-BWIM dataset (figshare, https://doi.org/10.7294/26880898.v1, CC0 license). The laboratory data, the processed data of the in-service bridge, the derived data behind every figure and table, and all analysis scripts with their console output are provided as Supplementary Material.

The manuscript is original, has not been published, and is not under consideration by any other journal. Two other papers by the authors are under review at *Structures*; they address different problems and share no data, test rig or text with this manuscript. All authors have approved the submission. Conflicts of interest, funding and the use of generative AI are declared in the manuscript.

Yours sincerely,

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